Proposed System

✅Uses “Machine Learning techniques”

1. Cloud Usage Prediction 📊

- The system analyzes historical cloud usage data (CPU, memory, traffic) using machine learning to forecast future resource requirements.

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Example: The system predicts that next week’s workload will need 50% more CPU and memory.

2. Idle and Underutilized Resource Detection 🔍

- The system minimizes cloud cost through idle resource detection, rightsizing, and ML-based predictive scaling.

Example: Before optimization: ₹10,000/month
After optimization: ₹6,200/month
Savings achieved: 38%

3. Cloud Cost Optimization 💰

- The system automatically adjusts cloud resources based on predictions and usage analysis to reduce overall cost.

Example: Idle virtual machines are shut down or resized to lower-cost instances.

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System Architecture

Flow Diagram (Top → Bottom)

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Phase 1

Cloud Usage Data 📊: Collect historical CPU, memory, traffic, and cost metrics.

Phase 2

Data Preprocessing ⚙️: Clean, normalize, and prepare data for ML models.

Phase 3

ML Prediction 🤖: Forecast future resource requirements using machine learning.

Phase 4

Cost Optimization 💰: Automatically resize or shut down resources to reduce cloud costs.

Existing System

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Cloud resources are managed “manually”.

Uses “basic monitoring dashboards”.

Fixed rule-based resource allocation.

No prediction of future usage.

#Results in “high cloud cost”.

Smart Cloud Cost optimization

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USING MACHINE LEARNING

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